

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4(a)    | <p>parental gametes      H    h    ×    H    h            ; <b>accept</b> from Punnett square</p> <p>offspring genotypes    HH    Hh    Hh    hh            ; <b>accept</b> from Punnett square</p> <p>offspring phenotypes        no disease        disease /        ;<br/>                                                                                 haemochromatosis</p>                                                                                                                                                                                                                                                                                                                                                     | 3 |
| 4(b)    | <p>any <b>two</b> from:</p> <ol style="list-style-type: none"> <li>1 reference to using knowledge of DNA sequence of, <i>HFE</i> gene / C282Y (mutation) ;</li> <li>2 (from) databases / bioinformatics ;</li> <li>3 (to) find / identify, the, <i>HFE</i> gene / C282Y (mutation), in fossil DNA sequences ;</li> <li>4 compare DNA sequence of fossil <i>HFE</i> gene with (sequence for), normal <i>HFE</i> allele / C282Y (mutation) ;</li> </ol>                                                                                                                                                                                                                                                                                 | 2 |
| 4(c)(i) | <p>any <b>four</b> from:</p> <ol style="list-style-type: none"> <li>1 low, iron / meat, (in diet), acts as a selection pressure ;</li> <li>2 people, with, C282Y / mutant allele, absorb more iron ;</li> <li>3 (so), survive / reproduce ;</li> <li>4 <i>idea that</i> (allele) frequency of mutant allele increased in the population ;</li> <li>5 natural selection ;</li> <li>6 (named) heterozygote advantage ; e.g. less likely to be iron deficient / not get haemochromatosis ;</li> <li>7 AVP ; e.g. C282Y (in heterozygotes) is neutral in present-day diets (so maintained)<br/>                                                                                 <b>or</b> gives protection from other diseases</li> </ol> | 4 |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(c)(ii) | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 highest / AW, percentage in Ireland where it first occurred ;</li> <li>2 higher percentages nearer Ireland due to migration ; <b>ora</b></li> <li>3 migration affected by, barriers / borders ;</li> <li>4 type of diet ;</li> <li>5 different treatments available (for hereditary haemochromatosis) in different countries ;</li> <li>6 AVP ;</li> </ol> | <b>3</b> |
| 4(d)     | any valid difference ; e.g. small database / ethnicity / region / location                                                                                                                                                                                                                                                                                                                                                        | <b>1</b> |
| 4(e)     | <p>282 ; ;</p> <p>if answer is incorrect <b>allow</b> 1 mark for correctly calculating the number of heterozygotes from incorrect 2pq<br/>i.e. <math>2501 \times \text{incorrect } 2pq</math></p>                                                                                                                                                                                                                                 | <b>2</b> |